



SIKSHA 'O' ANUSANDHAN

(Deemed to be University)

Faculty of Engineering & Technology (ITER)

Department of Computer Science and Engineering

Project Proposal Form

FINAL YEAR RESEARCH PROJECT-2026

SECTION:2241006

GROUP NO:01

PROJECT TITLE: Multi-Modal Disease Prediction Using Clinical and Laboratory Data with Explainable AI

PROJECT ABSTRACT:

Early detection of chronic diseases plays a crucial role in preventive healthcare and reducing long-term medical complications. However, most existing AI-based disease prediction models rely on a **single type of data**, such as only clinical parameters or only laboratory test values. This single-modal approach often leads to limited accuracy and poor generalization. To address this challenge, the proposed project aims to develop a **Multi-Modal Disease Prediction System** that integrates **clinical data** (age, gender, blood pressure, BMI, symptoms) with **laboratory data** (glucose, cholesterol, creatinine, hemoglobin, liver/kidney markers) to generate more accurate and reliable disease risk predictions.

The system will utilize machine learning models such as Random Forest, XGBoost, Logistic Regression, and Neural Networks to analyze multi-modal inputs. A multi-modal fusion layer will be applied to combine clinical and lab features, demonstrating improved predictive performance over traditional single-modal models. Additionally, the project incorporates **Explainable AI (XAI)** techniques such as SHAP and LIME to highlight the contribution of each feature in the final decision, ensuring transparency and interpretability for medical professionals.

The proposed system will be validated using real, publicly available medical datasets from Kaggle covering diseases such as diabetes, heart failure, kidney disease, and liver disorders. Evaluation metrics will include accuracy, precision, recall, F1-score, and ROC-AUC. The expected outcome is a robust and reliable AI model that can support early disease screening and assist healthcare providers in decision-making. This project delivers significant **social impact**, especially in resource-limited healthcare environments, by enabling fast, low-cost, and explainable disease risk assessment.

(1) SOFTWARE, HARDWARE OR METHODS/ALGORITHMS SPECIFICATIONS:

Software:

- **Python 3.10+, Jupyter Notebook**
- **Libraries: Pandas, NumPy, Scikit-learn, XGBoost, TensorFlow/PyTorch (optional), SHAP, Matplotlib**
- **Streamlit (for web interface)**
- **GitHub (version control)**

Hardware:

- **Standard laptop/PC with 8 GB RAM**
- **GPU optional but not required**

Methods / Algorithms Specifications :

○ **Learning Approach:**

Supervised Machine Learning and Ensemble Learning techniques applied on multimodal (clinical + laboratory) datasets for disease risk prediction.

○ **Data Preprocessing:**

Handling missing values, outlier removal, duplicate removal, categorical encoding, numerical feature scaling, normalization, multicollinearity check, and stratified train–test splitting.

○ **Feature Engineering:**

Segmentation of features into *Clinical Features* (age, gender, BP, BMI, symptoms) and *Laboratory Features* (glucose, creatinine, hemoglobin, cholesterol, liver/kidney markers). Correlation analysis, ANOVA tests, and feature importance ranking.

○ **Multi-Modal Fusion Framework:**

Fusion of clinical and laboratory inputs into a unified feature vector. Comparative analysis between single-modal and multi-modal prediction pipelines to demonstrate accuracy improvements.

○ **Machine Learning Algorithms:**

Logistic Regression, Support Vector Machine (SVM), Random Forest, Gradient Boosting, XGBoost, LightGBM, and Artificial Neural Networks (ANN).

○ **Class Imbalance Handling:**

Statistical resampling techniques including SMOTE, Oversampling, Undersampling, and Class-weight adjustments for balanced model training.

○ **Explainable AI Integration:**

Feature contribution analysis using SHAP (Shapley Additive Explanations) and LIME to interpret predictions and ensure transparency for medical decision support.

○ **Model Evaluation Pipeline:**

Accuracy, Precision, Recall, F1-Score, ROC-AUC, Confusion Matrix, and Comparative Performance analysis for all models across clinical-only, lab-only, and multi-modal configurations.

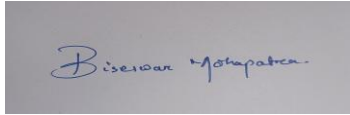
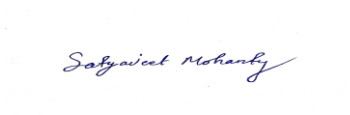
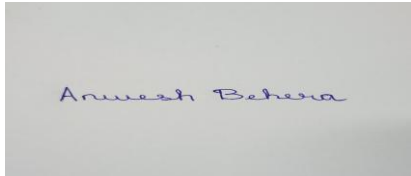
○ **Risk Prediction Strategy:**

Mapping predicted probabilities into Low, Medium, and High-risk categories to support early screening for diabetes, heart disease, kidney disease, and liver disease.

- **Hybrid Ensemble Strategy:**

Combining outputs of multiple high-performing models using soft voting / stacking ensemble techniques for optimized predictive performance.

Group Members with Signatures

Name	Registration No.	Signature
1) Biseswar Mohapatra	2241013319	
2) Satyajeeet Mohanty	2241013281	
3) Anwesh Behera	2241002065	
4) Rahul Mohanty	2241013265	

(2) **APPROVAL STATUS (To be filled in by the Section Coordinator of FRP):**

.....

Project Supervisor

.....

Section Coordinator, FRP

.....

FRP Coordinator